

Software Requirements Specification

for

Hospital Management System

Version 1.0 approved

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1. Introduction

1.1 Purpose

The product described in this Software Requirements Specification (SRS) is the Hospital Management System (HMS), Version 1.0, which represents the initial release of the system. The Hospital Management Application is a software application developed to enhance hospital operations and improve coordination between patients, medical staff, and administrators. The application provides a secure and efficient environment where users can perform essential hospital management tasks such as booking appointments, assigning staff to shifts, and accessing medical history or scheduling information. It will cover the core modules included in the first version of the hospital management.

1.2 Document Conventions

This Software Requirements Specification follows IEEE Std 830-1998 recommended practices for software requirements documentation. The following conventions have been applied throughout this document:

Requirement Identification

All functional requirements are identified using a standardized prefix format:

- REQ-[NUMBER]: Functional requirements within system features (e.g., REQ-1, REQ-2, REQ-3)
- REQ-[CATEGORY]-[NUMBER]: Requirements in Section 6 are prefixed by category (e.g., REQ-DB-1 for database requirements, REQ-LEGAL-1 for legal requirements)

Each requirement statement begins with "The system shall" to clearly indicate mandatory functionality.

Priority Inheritance

Priority ratings for system features (Section 4) are assigned at the feature level using four criteria: Benefit, Penalty, Cost, and Risk, each rated on a scale of 1-9. Individual functional requirements within each feature inherit the priority of their parent feature unless explicitly stated otherwise.

For example, all requirements under Feature 4.1 (Patient Account Management - Priority: High) are considered high priority requirements.

Text Formatting Conventions

- **Bold text:** Used for emphasis on key terms, requirement identifiers, user roles, and important system concepts
- **ALL CAPS:** Used only for established acronyms (e.g., HMS, API, HTTPS, FCM) defined in Appendix A

Placeholder Notation

TBD (To Be Determined): Indicates information or decisions that have not yet been finalized and are tracked in Appendix C. TBD items must be resolved during development sprints according to their assigned priority.

Section Numbering

This document follows hierarchical decimal numbering (e.g., 4.1.3, 6.2.1) to organize content logically and facilitate cross-referencing.

Terminology Consistency

All technical terms, acronyms, and domain-specific terminology are defined in Appendix A: Glossary to ensure consistent understanding across all stakeholders.

1.3 Intended Audience and Reading Suggestions

This document is intended for all stakeholders involved in developing and implementing the Hospital Management Application. It serves as a reference for developers to implement system functions, project managers to plan and track progress, testers to design test cases, and users and documentation writers to understand the system's capabilities.

The SRS is organized into sections that move from general to detailed information. The introduction outlines the purpose and scope, while later sections describe system features, functional, and non-functional requirements. Readers should begin with the Introduction and Overall Description to understand the system's purpose, followed by the Specific Requirements section for detailed technical information.

1.4 Product Scope

The purpose of the Hospital Management Application is to automate hospital operations by integrating various administrative, clinical, and patient-related processes into a single digital platform. The application aims to improve efficiency, accuracy, and accessibility in managing

appointments, patient records, staff scheduling, and other hospital services. By reducing manual paperwork and minimizing errors, the system enhances overall productivity and patient satisfaction. This software aligns with the hospital's strategic goal of digital transformation by providing a secure, user-friendly, and reliable system that supports better decision, faster service delivery, and improved coordination between medical staff and patients.

1.5 References

The following references were used in preparing this Software Requirements Specification (SRS) for the Hospital Management System (HMS):

1. **IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications**
Institute of Electrical and Electronics Engineers (IEEE), 1998.
2. **IEEE Std 29148-2018 – Systems and Software Engineering: Life Cycle Processes – Requirements Engineering**
International Organization for Standardization (ISO) / IEEE, 2018.
3. **Hospital Management System – Vision and Scope Document**
Version 1.0, Project Team, 2025.
4. **Use Case Model for Hospital Management System**
Internal Project Document, Version 1.0, 2025.
5. **UI/UX Design Guidelines for HMS Application**
Internal Design Document, Version 1.0, 2025.
6. **HL7 Healthcare Information Standards**
Health Level Seven International, <https://www.hl7.org/>.
7. **React Native Documentation**
Meta Platforms, <https://reactnative.dev/docs> (Accessed 2025).
8. **Node.js Documentation** (if used for backend)
OpenJS Foundation, <https://nodejs.org/en/docs> (Accessed 2025).

2. Overall Description

2.1 Product Perspective

The Hospital Management System (HMS) is a software product designed as a comprehensive digital solution for modern healthcare facilities. This system represents a complete replacement for traditional paper-based hospital management processes.

The HMS operates as a centralized platform that integrates multiple hospital departments and functions into a unified ecosystem. It serves as the primary interface between patients, medical staff, and hospital administrators, eliminating the need for separate systems for appointment scheduling, patient records, staff management, and administrative tasks.

The system architecture consists of a React Native cross-platform mobile and web application that communicates with Firebase cloud services for backend operations. This cloud-based approach ensures scalability, real-time data synchronization, and accessibility across multiple devices and platforms. The HMS interfaces with existing hospital infrastructure while maintaining data security and compliance requirements.

2.2 Product Functions

The Hospital Management System provides the following high-level functions :

- Patient Account Management: User registration, authentication, email verification, and password recovery services
- Appointment Scheduling: Booking, modification, and cancellation of patient appointments with medical staff
- Patient Records Management: Secure storage and retrieval of medical history, treatment plans, and diagnostic information
- Staff Scheduling and Assignment: Management of medical staff shifts, duty assignments, and availability tracking
- Real-time Notifications: Automated alerts for appointment confirmations, schedule changes, and system updates
- User Authentication and Authorization: Role-based access control for patients, doctors, staff, and administrators
- Data Synchronization: Real-time updates across all connected devices and user interfaces
- Cross-platform Accessibility: Support for Android, iOS

2.3 User Classes and Characteristics

The Hospital Management System serves three main user classes, each with specific characteristics and requirements :

Patients (Primary Users):

- Key characteristics: Need an intuitive interface, mobile accessibility, and secure access to personal health information
- Primary functions: Account registration, appointment booking, viewing medical history

Medical Staff/Doctors (Critical Users)

- Key characteristics: Require quick access to patient information, schedule management, and treatment updates
- Primary functions: Patient record management, schedule viewing, treatment documentation

Administrative Staff (Power Users)

- Key characteristics: Need comprehensive system access, reporting capabilities, and user management functions
- Primary functions: Staff scheduling, system administration, data management

The most critical user classes for system success are Patients and Medical Staff, as they represent the primary service providers and recipients in the healthcare environment

2.4 Operating Environment

The Hospital Management System (HMS) will operate in a simple environment suitable for small-scale applications.

Client Environment

- The system will run on mobile devices using:
 - **Android 8.0 or later**
 - **iOS 13 or later**
- The mobile application will be built using React Native, requiring:
 - Node.js (version 16+)

- Visual Studio Code (VS Code) as the primary code editor
- Android Studio (for Android builds)
- Xcode (for iOS builds, if needed)

Backend / Database

- **Database:**
- Firestore (cloud-based NoSQL database)
- No dedicated backend server required; Firestore handles data storage, authentication, and synchronization

Network Requirements

- Basic internet connection for:
 - Booking appointments
 - Viewing doctors and schedules
 - User authentication
- All communication with Firestore uses **HTTPS** for security

Hardware Requirements

- Any standard smartphone with:
 - At least 2 GB RAM
 - Basic storage availability (minimum 200 MB)

2.5 Design and Implementation Constraints

The development of the Hospital Management System (HMS) is subject to the following constraints that affect design and implementation choices:

1. Technology Constraints

- The mobile application must be developed using React Native to support both Android and iOS platforms.
- **Visual Studio Code (VS Code)** is the required development environment.
- **Node.js (version 16+ or later)** is required for the React Native CLI and package management.
- **Android Studio** and **Xcode** are required for platform-specific builds and testing.

- **Firestore** (Firebase) is the designated cloud database, providing data storage, authentication, and synchronization.

2. Hardware and Platform Constraints

- The application must run on standard smartphones with at least 2 GB RAM and 200 MB storage.
- The app must support Android 8.0+ and iOS 13+.

3. Interface and Integration Constraints

- The system must integrate with Firestore through the Firebase SDK.
- The app must communicate securely with Firestore using HTTPS.
- No other backend or server-side database will be used in this version.

4. Security Constraints

- User authentication must use Firebase Authentication to ensure secure login and password management.
- All sensitive data transmitted over the network must use encrypted communication (HTTPS).

5. Design and Programming Standards

- The code must follow consistent coding conventions for React Native and JavaScript/TypeScript.
- Comments and documentation must be included for maintainability.
- The delivered software may be maintained by the hospital's IT staff, so simplicity and readability are priorities.

6. Regulatory and Policy Constraints

- The application must comply with basic privacy standards to protect patient data.
- The system should minimize storage of sensitive data locally on devices.

2.6 User Documentation

The following documentation will be provided with the Hospital Management System (HMS):

- **User Manual:** A simple guide explaining how to sign up, log in, book or reschedule appointments, and view doctors. Provided as a PDF.
- **Online Help:** Short instructions and tips accessible within the app.
- **Quick Reference Guide:** A one-page summary of key tasks for easy use.

All documentation will use simple, clear language for patients and hospital staff.

2.7 Assumptions and Dependencies

Assumptions

- Users will have access to a smartphone with Android 8.0+ or iOS 13+.
- Users will have a basic internet connection to use the app features.
- Patients and hospital staff will be able to use the app without extensive technical knowledge.
- Firebase (Firestore) will remain available and reliable for database storage and authentication.

Dependencies

- The project depends on React Native for mobile app development.
- Node.js and VS Code are required for development.
- Android Studio and Xcode are needed for building and testing the app.
- Firestore SDK must be accessible and compatible with the React Native environment.

3. External Interface Requirements

3.1 User Interfaces

Hospital Management System (HMS) will provide a mobile user interface for patients and hospital staff. The logical characteristics of the interface are as follows:

User Interface Features

- Simple and clean screens for easy navigation on mobile devices.
- Standard buttons for actions such as **Submit**, **Cancel**, **Back**, and **Help**.
- Input fields for text, numbers, and selections with proper validation (e.g., age, phone number, email).
- Error messages displayed clearly below input fields if user input is invalid.
- Consistent layout across all screens with headers, menus, and action buttons in the same positions.
- Important information such as doctor names, appointment dates, and time slots displayed clearly and legibly.
- Help and guidance available via tooltips or a help button where necessary.

Screens / Components

The application will include user interfaces for the following main components:

1. **Sign-Up / Login Screens** – For user authentication and account creation.
2. **Home Screen** – Overview of available actions for patients.
3. **Appointments Screen** – Viewing, booking, and rescheduling appointments.
4. **Profile Screen** – Viewing and editing personal information.
5. **Forgot Password Screen** – Recovering a lost password.
6. **Dashboard Screen** – Overview of user activity, notifications, and key information.
7. **Chat Screen** – Messaging between patients and hospital staff or doctors.
8. **Clinics Screen** – Viewing available clinics and their services.

3.2 Hardware Interfaces

The Hospital Management System Application is designed to operate on a variety of hardware platforms, including Android and iOS smartphones, tablets, laptops, and desktop computers. The system requires devices with standard input and output capabilities, such as touchscreens, keyboards, and display units, to enable user interaction with the application's graphical interface.

The application depends on the device's internet connectivity (via Wi-Fi or mobile data) to communicate with cloud-based services for data storage, authentication, and real-time updates. All hardware-level interactions follow standard mobile and desktop operating system communication protocols to ensure consistent performance across devices.

Data and control signals between the device and the cloud are transmitted through secure HTTPS protocols, ensuring encrypted and reliable communication between the client device and the server.

3.3 Software Interfaces

The Hospital Management Application interacts with several external software components that support its core functionality and data management. The application connects to Firebase services for secure authentication, real-time data storage, and cloud communication. Specifically, it interfaces with Firebase Authentication to manage user registration, email verification, and login processes, ensuring only authorized users can access hospital services.

It also integrates with Cloud Firestore, which serves as the main cloud database for storing and retrieving patient records, appointment information, and staff schedules. Data flows between the application and Firebase services through secure HTTPS protocols, ensuring encrypted communication and data integrity. Input data, such as user credentials, patient details, and appointment requests, are transmitted to Firestore, while output data, such as appointment confirmations or medical record details, are retrieved and displayed to the user.

Additionally, the system uses Firebase Cloud Messaging (FCM) to deliver real-time notifications, such as appointment confirmations or schedule changes, to patients and hospital staff. All communications between the system and these external software components must comply with data privacy and security standards to ensure the confidentiality and integrity of sensitive information the hospital has.

3.4 Communications Interfaces

The Hospital Management System Application relies on secure internet-based communication to exchange data between the client devices and the cloud server. All communication occurs over HTTPS (Hypertext Transfer Protocol Secure), which ensures encrypted data transfer and protects against unauthorized access or interception. The system communicates with Firebase cloud services, including Firestore for real-time database operations and Firebase Authentication for secure user login and registration. Data transmitted includes patient details, appointment requests, staff schedules, and confirmation messages. The responses from the server provide updated information to the user interface in real time.

During the registration process, the system supports email-based communication through Firebase Authentication, which automatically sends a verification email to newly registered users to confirm their identity before granting full access to the application. The system must maintain continuous synchronization between client devices and cloud data. Any update performed by a user (such as booking an appointment or modifying a schedule) will be immediately reflected in the database and available to other authorized users.

4. System Features

4.1 Patient Account Management

4.1.1 Description and Priority

This feature enables patients to register, authenticate, and manage their access to the hospital's online system securely. It includes account creation, email verification, login, and password recovery functionalities.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 8
- Cost: 5
- Risk: 4

4.1.2 Stimulus/Response Sequences

Signup Sequence:

- **Stimulus:** The patient selects "Create New Account" and enters required personal and contact information.
- **Response:** The system validates inputs, stores user details, and sends a verification email to the patient.
- **Login Sequence:**
- **Stimulus:** The patient enters valid credentials on the login page.
- **Response:** The system authenticates the credentials and grants access to the patient dashboard.
- **Forgot Password Sequence:**
- **Stimulus:** The patient selects "Forgot Password" and enters their registered email.
- **Response:** The system sends a password reset link via email and allows the user to set a new password upon verification.
- **Email Verification Sequence:**
- **Stimulus:** The patient clicks the verification link sent to their email after signup.
- **Response:** The system confirms the link, activates the account, and notifies the user of successful verification.

4.1.3 Functional Requirements

REQ-1: The system shall allow patients to create a new account by providing personal details (e.g., name, date of birth, gender) and contact information (email, phone number).

REQ-2: The system shall validate all input fields and display appropriate error messages for missing or invalid information.

REQ-3: Upon successful registration, the system shall send an authentication email containing a verification link to the patient's provided email address.

REQ-4: The system shall activate a patient's account only after successful email verification.

REQ-5: The system shall provide a secure login page where patients can enter their registered email and password to access their account.

REQ-6: The system shall deny access and display an error message if login credentials are invalid.

REQ-7: The system shall provide a "Forgot Password" option allowing patients to reset their password via a secure email link.

REQ-8: The system shall ensure that password reset links expire after a defined time period (TBD).

REQ-9: The system shall encrypt all passwords and sensitive information before storing them in the database.

4.2 Appointment Management

4.2.1 Description and Priority

This feature allows patients to manage their medical appointments efficiently, doctors to view their schedules, and receptionists to oversee clinic-wide bookings.

Patients can browse clinics, select doctors, book, reschedule, and cancel appointments. Doctors can view their daily and upcoming schedules, while receptionists can monitor and adjust appointments for all doctors.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 9
- Cost: 7
- Risk: 5

4.2.2 Stimulus/Response Sequences

Clinic and Doctor Selection Sequence:

Stimulus: The patient selects “View Clinics” and chooses a specific clinic.

Response: The system displays available doctors within the selected clinic, along with their time slots.

Appointment Booking Sequence:

Stimulus: The patient selects a doctor and time slot, then confirms the booking.

Response: The system validates slot availability, reserves it, and displays a confirmation message to the patient.

View Appointments Sequence:

Stimulus: The patient opens the “My Appointments” page.

Response: The system retrieves and displays both upcoming and past appointments.

Reschedule Appointment Sequence:

Stimulus: The patient selects an existing appointment and requests a new time slot.

Response: The system verifies availability, updates the appointment, and notifies the doctor of the change.

Cancel Appointment Sequence:

Stimulus: The patient chooses to cancel a booked appointment.

Response: The system confirms cancellation and frees the slot for other patients.

Doctor View Schedule Sequence:

Stimulus: The doctor logs in and opens “My Schedule.”

Response: The system displays all upcoming appointments, including patient details and time slots.

Receptionist View and Manage Appointments Sequence:

Stimulus: The receptionist opens “Clinic Appointments.”

Response: The system retrieves all appointments across doctors, enabling the receptionist to view and assist with scheduling or modifications when required.

4.2.3 Functional Requirements

REQ-1: The system shall allow patients to browse clinics and view available doctors and their time slots.

REQ-2: The system shall enable patients to book, reschedule, and cancel appointments.

REQ-3: The system shall display upcoming and past appointments to patients.

REQ-4: The system shall allow doctors to view their scheduled appointments and associated patient information.

REQ-5: The system shall provide receptionists access to all doctors’ schedules to assist in managing appointments.

REQ-6: The system shall prevent double-booking of time slots.

REQ-7: The system shall notify relevant users (doctor, patient, receptionist) of appointment updates or cancellations.

REQ-8: The system shall validate input data and handle invalid requests gracefully (e.g., invalid time, unavailable slot).

4.3 Staff Account Management

4.3.1 Description and Priority

This feature allows administrators to manage staff (doctors, nurses, and receptionists) accounts and ensures secure access for all employees. The admin can create, deactivate, or delete staff accounts, while staff members can log in and recover their passwords if forgotten.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 8
- Cost: 5
- Risk: 4

4.3.2 Stimulus/Response Sequences

Staff Account Creation Sequence:

Stimulus: The admin selects “Create New Staff Account” and enters the required staff details (name, role, contact information, and credentials).

Response: The system validates the data, creates the account, and sends login credentials to the staff member’s email.

Account Deactivation or Deletion Sequence:

Stimulus: The admin selects a staff account and chooses to deactivate or delete it.

Response: The system verifies admin authorization, updates the account status, and restricts the staff member from further access.

Staff Login Sequence:

Stimulus: A staff member enters valid credentials on the login page.

Response: The system authenticates the credentials and grants access to the staff dashboard based on role (doctor, nurse, or receptionist).

Forgot Password Sequence:

Stimulus: A staff member selects “Forgot Password” and enters their registered email.

Response: The system sends a password reset link via email and allows the staff member to set a new password upon verification.

4.3.3 Functional Requirements

REQ-1: The system shall allow administrators to create staff accounts with appropriate roles and permissions.

REQ-2: The system shall allow administrators to deactivate or permanently delete staff accounts.

REQ-3: The system shall allow staff members to log in using their assigned credentials.

REQ-4: The system shall provide a “Forgot Password” feature to reset passwords securely.

REQ-5: The system shall send email notifications to staff upon account creation, deactivation, or password reset.

REQ-6: The system shall ensure only authorized admins can manage staff accounts.

REQ-7: The system shall validate inputs (e.g., unique email, required fields) and handle errors gracefully.

4.4 Patient Medical Record Management

4.4.1 Description and Priority

This feature enables the creation, updating, and secure storage of patient medical records. Patients can fill in initial medical information, doctors can update these records after each visit, and administrators ensure that all records are securely stored and accessible only to authorized users.

Priority: Medium

Priority Ratings:

- Benefit: 7
- Penalty: 7
- Cost: 6
- Risk: 5

4.4.2 Stimulus/Response Sequences

Patient Record Creation Sequence:

Stimulus: The patient fills out a medical form with personal and health-related information during account setup or before their first appointment.

Response: The system validates the input and securely stores the data as the patient's medical record.

Doctor Record Update Sequence:

Stimulus: After a patient's visit, the doctor accesses the patient's record and updates it with diagnosis, treatment details, and prescribed medications.

Response: The system saves the updated record and maintains version control to ensure data integrity.

Secure Storage Sequence:

Stimulus: The administrator configures storage policies and system permissions for medical records.

Response: The system encrypts all records and ensures that only authorized users (doctors, patients, and admins) can access relevant data.

4.4.3 Functional Requirements

REQ-1: The system shall allow patients to create and submit their initial medical records through a form.

REQ-2: The system shall allow doctors to view and update patient medical records after each visit.

REQ-3: The system shall maintain a complete history of updates made to patient records.

REQ-4: The system shall restrict access to medical records based on user roles (patient, doctor, admin).

REQ-5: The system shall securely store all medical records using encryption.

REQ-6: The system shall comply with privacy and confidentiality standards for patient data.

REQ-7: The system shall validate all entered medical data before storing or updating records.

REQ-8: The system shall notify the patient when their record is updated by a doctor.

4.5 Shift Scheduling Management

4.5.1 Description and Priority

This feature enables the hospital administration to manage work schedules for nurses. Administrators can create and edit available shifts, assign nurses to these shifts, and view the full schedule. Nurses can mark unavailable days to avoid being assigned during those times and can view their assigned shifts on a schedule page.

Priority: Medium

Priority Ratings:

- Benefit: 7
- Penalty: 6
- Cost: 5
- Risk: 4

4.5.2 Stimulus/Response Sequences

Shift Creation Sequence:

Stimulus: The admin opens the shift management page and creates new shifts by specifying start time, end time, and required staff count.

Response: The system stores the new shift details in the database and makes them available for assignment.

Nurse Availability Sequence:

Stimulus: A nurse accesses their schedule settings and marks specific days as unavailable.

Response: The system updates the nurse's profile to exclude those dates from assignment options.

Shift Assignment Sequence:

Stimulus: The admin assigns nurses to available shifts.

Response: The system updates the schedule, notifies the assigned nurses, and displays their updated shift calendar.

Schedule Viewing Sequence:

Stimulus: A nurse logs in and opens the schedule page.

Response: The system retrieves and displays all assigned shifts for the nurse in a calendar format.

4.5.3 Functional Requirements

REQ-1: The system shall allow administrators to create, edit, and delete available shifts.

REQ-2: The system shall allow nurses to mark unavailable days on their schedules.

REQ-3: The system shall allow administrators to assign nurses to specific shifts.

REQ-4: The system shall display assigned shifts to nurses on their schedule page.

REQ-5: The system shall send a notification to nurses upon assignment or change in schedule.

REQ-6: The system shall prevent assigning a nurse to overlapping shifts or shifts on unavailable days.

REQ-7: The system shall allow the admin to view all nurses' schedules in an aggregated view.

REQ-8: The system shall store all scheduling data securely and update it in real time.

4.6 Administrative Dashboard and Monitoring

4.6.1 Description and Priority

This feature provides the hospital administrator with a centralized dashboard for system oversight. It allows viewing user activity reports, exporting them for management purposes, monitoring system performance metrics, and receiving alerts about abnormal or suspicious activities. This feature enhances operational efficiency and helps maintain system stability and security.

Priority: Medium

Priority Ratings:

- Benefit: 7
- Penalty: 5
- Cost: 5
- Risk: 4

4.6.2 Stimulus/Response Sequences

Dashboard Access Sequence:

Stimulus: The admin logs into the system and selects the dashboard option.

Response: The system retrieves and displays reports summarizing user activity, usage statistics, and system performance metrics.

Report Export Sequence:

Stimulus: The admin selects the option to export reports in a chosen format (e.g., PDF, CSV).

Response: The system generates the requested report and provides a downloadable file.

Performance Monitoring Sequence:

Stimulus: The admin views the “System Performance” section to check for slow queries, server usage, or uptime.

Response: The system displays real-time performance data and trends.

Abnormal Activity Alert Sequence:

Stimulus: The system detects unusual patterns (e.g., multiple failed login attempts, sudden traffic spikes).

Response: The system sends an alert or notification to the admin dashboard and optionally via email.

4.6.3 Functional Requirements

REQ-1: The system shall provide an admin dashboard displaying real-time user and system statistics.

REQ-2: The system shall allow administrators to export reports in multiple formats (e.g., PDF, CSV).

REQ-3: The system shall monitor system performance metrics such as response time, uptime, and database load.

REQ-4: The system shall automatically detect and log abnormal activities or security anomalies.

REQ-5: The system shall notify administrators immediately when abnormal activity is detected.

REQ-6: The system shall allow filtering of dashboard reports by user role, date, or activity type.

REQ-7: The system shall ensure that only administrators can access dashboard data and performance logs.

4.7 Patient Communication and Support

4.7.1 Description and Priority

This feature enables patients to communicate directly with the hospital's support team and provide feedback after their appointments. It allows patients to send inquiries or complaints without needing to visit in person and enables support staff to view and respond to these messages efficiently. The feature also supports collecting patient ratings and feedback for service improvement.

Priority: Low

Priority Ratings:

- Benefit: 5
- Penalty: 3
- Cost: 5
- Risk: 3

4.7.2 Stimulus/Response Sequences

Messaging Sequence:

Stimulus: The patient navigates to the "Support" page and submits a message or inquiry.

Response: The system records the message and notifies the support group. The support member can view the message and send a reply, which appears in the patient's inbox.

Feedback Sequence:

Stimulus: After completing an appointment, the patient submits feedback or a rating.

Response: The system stores the feedback and makes it available to administrators for quality assessment and reporting.

4.7.3 Functional Requirements

REQ-1: The system shall allow patients to submit messages or inquiries to hospital support.

REQ-2: The system shall notify support staff when a new message is received.

REQ-3: The system shall allow support staff to view, reply to, and close patient inquiries.

REQ-4: The system shall allow patients to submit feedback or ratings after appointments.

REQ-5: The system shall store all communication and feedback securely for administrative review.

4.8 Administrative Dashboard and Monitoring

4.8.1 Description and Priority

This feature allows patients to view their medical bills, make online payments, and request refunds for canceled appointments. It streamlines financial transactions between patients and the hospital, reducing manual handling and improving convenience.

Priority: Medium

Priority Ratings:

- Benefit: 7
- Penalty: 6
- Cost: 6
- Risk: 9

4.8.2 Stimulus/Response Sequences

View Bill Sequence:

Stimulus: The patient navigates to the “Billing” section in the system to check pending or paid invoices.

Response: The system retrieves and displays a detailed list of the patient’s bills, including payment status and breakdown of charges.

Payment Sequence:

Stimulus: The patient selects an unpaid bill and chooses a payment method (e.g., credit card, e-wallet).

Response: The system processes the payment securely and updates the payment status to “Paid.” A digital receipt is generated and sent to the patient.

Refund Sequence:

Stimulus: The patient cancels an appointment and requests a refund through the billing page.

Response: The system verifies eligibility for refund, processes the transaction, and updates the billing record to reflect the refund status.

4.8.3 Functional Requirements

REQ-1: The system shall allow patients to view their medical bills and payment history.

REQ-2: The system shall enable patients to make secure payments through integrated payment gateways.

REQ-3: The system shall generate and send digital receipts upon successful payments.

REQ-4: The system shall support refund requests for canceled appointments.

REQ-5: The system shall update billing records automatically after successful payment or refund.

REQ-6: The system shall ensure all financial transactions are securely processed and logged for auditing.

4.9 Administrative Dashboard and Monitoring

4.9.1 Description and Priority

This feature allows administrators and doctors to view, set, and manage doctors' schedules efficiently. It supports assigning shifts, defining weekly working hours, and updating or canceling availability to ensure accurate appointment booking and department coverage.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 8
- Cost: 6
- Risk: 7

4.9.2 Stimulus/Response Sequences

Schedule Viewing Sequence:

Stimulus: The admin/scheduler navigates to the "Doctor Schedules" page.

Response: The system displays all doctors' schedules, including their assigned shifts and current availability.

Schedule Update Sequence:

Stimulus: A doctor selects their profile and chooses to update or cancel a scheduled shift.

Response: The system updates the doctor's schedule in real-time and notifies affected appointments or administrators if applicable.

Availability Setup Sequence:

Stimulus: A doctor opens the "Set Availability" section and defines their weekly working hours.

Response: The system records the specified availability, which becomes visible to the appointment booking module for patients.

4.9.3 Functional Requirements

REQ-1: The system shall allow administrators to view all doctors' schedules filtered by department, date, or doctor name.

REQ-2: The system shall allow doctors to set their weekly working hours and specify available time slots.

REQ-3: The system shall allow doctors to update or cancel scheduled shifts when their availability changes.

REQ-4: The system shall automatically update patient appointment availability based on doctors' schedule changes.

REQ-5: The system shall notify the admin/scheduler when a doctor modifies their schedule to adjust coverage if necessary.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

I. System Response Time

- The system shall respond to user actions (such as login, page navigation, or form submission) within 3 seconds under normal network conditions.
- Appointment booking confirmation, payment processing, and feedback submission shall be completed within 3 seconds after the user submits the request.
- Search results for clinics, doctors, or appointments shall be displayed within 2 seconds for up to 1,000 records.

II. Concurrent Users and Load Handling

- The system shall support a minimum of 500 concurrent users (patients, doctors, staff, and admins) without noticeable degradation in performance.
- The system shall handle up to 50 concurrent appointment bookings per second without data conflicts or system slowdown.

III. Data Processing and Storage

- Database queries shall execute within 1 second for common operations such as retrieving user profiles, appointments, or medical records.
- For the appointment management features, data synchronization between patient and doctor schedules shall be completed within 2 seconds of booking, canceling, or rescheduling.

IV. Data availability

- Patient and hospital data shall be accessible at all times from redundant data storage systems.
- Regular database backups shall occur at least once every 6 hours to prevent data loss during failures.

5.2 Safety Requirements

<Specify those requirements that are concerned with possible loss, damage, or harm that could result from the use of the product. Define any safeguards or actions that must be taken, as well as actions that must be prevented. Refer to any external policies or regulations that state safety issues that affect the product's design or use. Define any safety certifications that must be satisfied.>

I. Data Integrity Protection

- The system shall ensure that no patient data (medical records, appointment data, or billing information) is lost or corrupted during operations, system crashes, or power failures.
-

5.3 Security Requirements

<Specify any requirements regarding security or privacy issues surrounding use of the product or protection of the data used or created by the product. Define any user identity authentication requirements. Refer to any external policies or regulations containing security issues that affect the product. Define any security or privacy certifications that must be satisfied.>

I. Authentication and Authorization

- Every user (patient, doctor, nurse, admin, or support staff) must log in using unique credentials.
- Role-based access control (RBAC) shall be implemented to ensure users can only access data and features relevant to their role.
- Password input fields shall be masked (e.g., shown as dots or asterisks) during entry to prevent shoulder surfing or visual exposure of sensitive data.
- The system shall provide an optional “Show Password” toggle, allowing users to verify their input securely if needed.

5.4 Software Quality Attributes

The Hospital Management System (HMS) will be designed to ensure high usability, reliability, performance, security, maintainability, portability, and testability. The application will be simple and intuitive for both patients and hospital staff, allowing key actions such as booking appointments or viewing doctors to be completed in a few taps, with clear guidance and error messages where necessary. Reliability is a priority, and the system is expected to be available 99% of the time during operational hours, with all data stored in Firestore remaining consistent and accessible. Performance targets include screens loading within two seconds on standard smartphones and near real-time synchronization with the database. Security measures include the use of Firebase Authentication for user login, HTTPS for all communication, and encryption of sensitive data stored locally when needed. The application will follow consistent coding standards and include comments to ensure maintainability, making it easy to update or extend with new features in future releases. Portability is ensured by supporting Android 8.0+ and iOS 13+ devices across various screen sizes. Finally, the system will be testable through unit tests, functional testing, and user acceptance testing, with proper input validation to facilitate both automated and manual testing.

5.5 Business Rules

The Hospital Management System (HMS) will operate under a set of business rules that define how users interact with the system and what actions they are allowed to perform. Only registered patients can book, reschedule, or view their own appointments, while doctors and administrative staff have access to view schedules and manage clinic-related information. Patients are responsible for providing accurate personal information during sign-up, including age, phone number, and email, which must be validated by the system. The system enforces secure access, so only authenticated users can access sensitive data, and different roles have different permissions to ensure privacy and compliance. Appointments can only be booked within available time slots provided by doctors or clinics, and the system automatically prevents double booking. Any changes to schedules or appointments must be reflected in real-time to maintain accuracy across all users. These business rules guide the system's functionality and ensure orderly, secure, and efficient operation.

6. Other Requirements

6.1 Database Requirements

Data Storage and Management

The Hospital Management System shall utilize Firestore (Cloud Firestore) as the primary database for all data persistence and real-time synchronization. All patient records, appointment information, staff schedules, billing details, and system logs shall be stored in structured collections within Firestore.

Specific Database Requirements:

- **REQ-DB-1:** The system shall store all data in Firestore collections with appropriate indexing to optimize query performance for frequently accessed data (patients, appointments, doctors, schedules).
- **REQ-DB-2:** The system shall maintain data relationships using document references and subcollections to ensure proper data normalization and consistency.
- **REQ-DB-3:** The system shall implement automated backup procedures to create copies of all critical data at least every 6 hours.
- **REQ-DB-4:** The system shall maintain audit logs for all data modifications, including timestamps, user identifiers, and the nature of changes for compliance and troubleshooting purposes.
- **REQ-DB-5:** The system shall implement data retention policies defining how long historical records (e.g., past appointments, archived communications) are maintained before archival or deletion.
- **REQ-DB-6:** The system shall ensure database scalability to accommodate growth in users, appointments, and data volume over time without performance degradation.

6.2 Internationalization Requirements

Multi-Language Support

While the initial release (Version 1.0) will be deployed in English, the system architecture shall support future internationalization and localization efforts.

Internationalization Requirements:

- **REQ-I18N-1:** The system shall be designed with language-agnostic architecture, allowing easy addition of new languages in future sprints.
- **REQ-I18N-2:** The system shall support date, time, and number formatting according to locale-specific standards when internationalization is implemented.

6.3 Legal and Regulatory Requirements

Compliance Standards

The Hospital Management System must comply with healthcare regulations and data privacy standards applicable to hospital operations in Egypt and internationally.

Legal and Regulatory Requirements:

- **REQ-LEGAL-1:** The system shall comply with the General Data Protection Regulation (GDPR) for patients with European data origins, ensuring data subject rights and privacy controls.
- **REQ-LEGAL-2:** The system shall adhere to basic healthcare information security standards and protect patient information from unauthorized access or disclosure.
- **REQ-LEGAL-3:** The system shall maintain compliance with data protection regulations regarding storage, processing, and transmission of sensitive patient medical records.
- **REQ-LEGAL-4:** The system shall implement privacy policies and terms of service that inform users about data collection, usage, and storage practices.
- **REQ-LEGAL-5:** All user agreements and system documentation shall include explicit consent mechanisms for data processing and third-party sharing (if applicable).
- **REQ-LEGAL-6:** The system shall maintain comprehensive audit trails and logs to demonstrate compliance with regulatory requirements during audits or investigations.
- **REQ-LEGAL-7:** The system shall provide mechanisms for users to request, access, and delete their personal data in compliance with privacy regulations.

6.4 Reusability Objectives

Code and Component Reusability

The development team shall design the system with reusability in mind to minimize technical debt and facilitate future enhancements.

Reusability Objectives:

- **REQ-REUSE-1:** Common UI components (buttons, input fields, modal dialogs, date pickers) shall be built as reusable React Native components with configurable props and consistent styling.
- **REQ-REUSE-2:** Utility functions for data validation, date/time formatting, and string manipulation shall be centralized in a utilities module for use across all features.
- **REQ-REUSE-3:** Firebase integration logic (authentication, Firestore queries, real-time listeners) shall be abstracted into reusable services and utility modules to reduce code duplication.
- **REQ-REUSE-4:** Authentication and authorization logic shall be implemented as a reusable middleware service that can be applied to multiple routes and features.
- **REQ-REUSE-5:** Error handling and logging mechanisms shall be centralized and reusable across all modules and screens.
- **REQ-REUSE-6:** API communication patterns and request/response handling shall follow consistent, reusable conventions throughout the application.

Appendix A: Glossary

This glossary defines key terms, acronyms, and abbreviations used throughout the Hospital Management System Software Requirements Specification document.

Technical Terms

API (Application Programming Interface): A set of protocols and tools that allows different software applications to communicate with each other.

Authentication: The process of verifying the identity of a user, device, or system before granting access to resources.

Authorization: The process of determining what permissions or access rights an authenticated user has within the system.

Backend: The server-side components of the application responsible for data processing, business logic, and database operations.

CI/CD (Continuous Integration/Continuous Deployment): Automated processes for building, testing, and deploying software changes.

Cloud Firestore: A flexible, scalable NoSQL cloud database provided by Firebase for storing and syncing data in real-time.

Encryption: The process of converting data into a coded format to prevent unauthorized access during storage or transmission.

FCM (Firebase Cloud Messaging): A cross-platform messaging solution that enables the delivery of notifications and messages to mobile devices.

Firebase: A comprehensive mobile and web application development platform provided by Google that includes authentication, database, cloud storage, and messaging services.

Firebase Authentication: A Firebase service that provides backend services to authenticate users using passwords, phone numbers, and federated identity providers.

Frontend: The client-side components of the application that users interact with directly through the user interface.

HTTPS (Hypertext Transfer Protocol Secure): A secure version of HTTP that encrypts data transmitted between client and server.

NoSQL Database: A non-relational database that provides flexible schemas and is designed for distributed data stores.

React Native: An open-source mobile application framework for building native mobile apps using JavaScript and React.

Real-time Synchronization: The automatic and immediate updating of data across all connected devices and user interfaces.

SDK (Software Development Kit): A collection of software development tools and libraries that enable developers to create applications for specific platforms.

TBD (To Be Determined): A placeholder indicating that specific details or decisions have not yet been finalized.

UI/UX (User Interface/User Experience): The visual design and overall experience a user has while interacting with the application.

Healthcare and Hospital Terms

Administrator: A hospital staff member with full system access responsible for managing users, configurations, and system operations.

Appointment: A scheduled meeting between a patient and a healthcare provider (doctor, nurse, specialist) at a specific date and time.

Clinic: A healthcare facility or department within the hospital that specializes in treating specific conditions or providing particular services.

Doctor: A licensed medical professional who diagnoses, treats, and provides medical care to patients.

Electronic Health Record (EHR): A digital version of a patient's medical history, treatment plans, diagnoses, and medications.

HMS (Hospital Management System): The software application being developed to manage hospital operations, patient care, and administrative tasks.

Medical History: A record of a patient's past medical conditions, treatments, surgeries, medications, and health-related information.

Medical Record: A comprehensive document containing a patient's health information, including diagnoses, treatments, test results, and prescriptions.

Nurse: A healthcare professional who provides patient care, administers medications, and assists doctors in medical procedures.

Patient: An individual who receives medical care or treatment from healthcare providers within the hospital system.

Receptionist: A staff member responsible for managing appointments, greeting patients, and handling administrative tasks at the hospital front desk.

Shift: A scheduled work period assigned to nurses or staff members during which they are responsible for patient care or hospital operations.

Stakeholder: Any individual or group with an interest in the Hospital Management System, including patients, doctors, nurses, administrators, and hospital management.

Regulatory and Compliance Terms

Audit Trail: A chronological record of system activities and data changes used for security, compliance, and troubleshooting purposes.

Compliance: Adherence to laws, regulations, standards, and policies governing healthcare data management and patient privacy.

Data Privacy: The protection of personal and sensitive information from unauthorized access, use, or disclosure.

GDPR (General Data Protection Regulation): A European Union regulation governing data protection and privacy for individuals within the EU and European Economic Area.

HL7 (Health Level Seven): A set of international standards for the exchange, integration, and retrieval of electronic health information.

System-Specific Terms

Availability Status: The current state indicating whether a doctor, nurse, or time slot is available for appointments or shift assignments.

Dashboard: A centralized interface displaying key information, metrics, and system activities for administrators or users.

Double-booking: An error condition where two or more appointments are scheduled for the same doctor at the same time slot.

Email Verification: A security process that confirms a user's email address by sending a verification link that must be clicked to activate the account.

Password Reset: A security feature allowing users to create a new password when they have forgotten or wish to change their existing password.

Role-Based Access Control (RBAC): A security model that restricts system access based on a user's assigned role (patient, doctor, nurse, receptionist, administrator).

Session: A temporary period during which a user is logged into the system and has active access to their account features.

Time Slot: A specific time interval during which a doctor is available to see patients for appointments.

User Story: A short description of a feature or functionality from the perspective of an end-user, used in agile development.

Appendix C: To Be Determined List

TBD-001: Password Reset Link Expiration Time

Section Reference: 4.1.3 (Patient Account Management - REQ-8)

Description: The specific expiration time for password reset links has not been determined.

Options Under Consideration:

- 15 minutes
- 30 minutes
- 1 hour
- 24 hours

Priority: High

Impact: Affects user security and convenience balance

Notes: Industry standard is typically 15-30 minutes for security purposes

TBD-002: Concurrent User Capacity

Section Reference: 5.1 (Performance Requirements)

Description: Maximum number of concurrent users the system should support needs to be specified.

Options Under Consideration:

- 100 concurrent users (small hospital)
- 500 concurrent users (medium hospital)
- 1000+ concurrent users (large hospital)

Priority: Medium

Impact: Influences infrastructure planning and load testing requirements

Notes: Based on expected hospital size and patient volume

TBD-003: Payment Gateway Integration

Section Reference: 4.8 (Billing and Payment Processing)

Description: Specific payment gateway provider and integration method to be determined.

Options Under Consideration:

- PayPal
- Local Egyptian payment gateway (Fawry, Paymob)
- Multiple gateway support

Priority: High

Impact: Critical for billing feature implementation

Notes: Must support local payment methods popular in Egypt

TBD-004: Notification Delivery Channels

Section Reference: 2.2 (Product Functions), 3.4 (Communications Interfaces)

Description: Additional notification channels beyond Firebase Cloud Messaging need to be determined.

Channels to Consider:

- SMS text messages for critical alerts
- Email notifications for appointments
- In-app notifications only
- WhatsApp integration

Priority: Medium

Impact: Affects user engagement and appointment attendance rates

Notes: SMS may incur additional costs; evaluate cost-benefit

TBD-005: Multi-language Support Priority

Section Reference: 6.2 (Internationalization Requirements)

Description: Priority and timeline for implementing Arabic language support need to be determined.

Options:

- Sprint 3 (immediate next priority)
- Sprint 4-5 (medium-term)
- Post Version 1.0 release

Priority: Medium

Impact: Significant for Egyptian market adoption

Notes: Arabic is essential for broader hospital staff and patient accessibility

TBD-006: Appointment Reminder Timing

Section Reference: 4.2 (Appointment Management)

Description: When and how frequently to send appointment reminders to patients.

Options Under Consideration:

- 24 hours before appointment
- 24 hours + 1 hour before appointment

- 48 hours + 24 hours + 1 hour before appointment
- User-configurable reminder preferences

Priority: Medium

Impact: Reduces no-show rates and improves patient satisfaction

Notes: Multiple reminders may improve attendance but could annoy users

TBD-007: Offline Mode Capabilities

Section Reference: 2.7 (Assumptions and Dependencies)

Description: Determine what functionality, if any, should be available when the device is offline.

Options:

- No offline support (current assumption)
- View-only access to cached data
- Full offline mode with sync when reconnected

Priority: Low

Impact: Significant development effort; affects user experience in poor connectivity

Notes: Firebase supports offline persistence, but requires careful implementation